📝 Conservation of Mass — Practice Problems

Grade 8 Science · Chemical Change & the Mass of Substances

Grade ______ Class ______ No. ______ Name ____________________

Part 1 What happens to the mass? 5 patterns
Picture each experiment and choose

1. **Copper powder** on a combustion dish was heated **in open air**. What happens to the mass?

Answer **A**

A **Increases**　**B：Decreases**　**C：Stays the same**

> 💡 It gets heavier by exactly the amount of **oxygen from the air that bonds with the copper** (copper + oxygen → copper oxide). Same mechanism as the steel wool (iron) in the simulation.

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2. Copper was placed in a round-bottom flask, **sealed with a rubber stopper**, and heated. What happens to the mass?

Answer **C**

A：Increases　**B：Decreases**　**C** **Stays the same**

> 💡 The oxygen that bonds with the copper is **already inside the flask from the start**. Since nothing enters or leaves, the total mass stays the same.

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3. Dilute sulfuric acid was mixed with barium hydroxide solution, and a **white precipitate** (barium sulfate) formed. What happens to the mass?

Answer **C**

A：Increases　**B：Decreases**　**C** **Stays the same**

> 💡 No gas is produced in this reaction. Only the **combination of atoms changes**, so the mass stays the same even without a lid.

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4. Inside **a sealed jar**, hydrochloric acid was added to sodium bicarbonate, and carbon dioxide was produced. What happens to the mass?

Answer **C**

A：Increases　**B：Decreases**　**C** **Stays the same**

> 💡 The carbon dioxide that was produced is also **trapped inside the jar**, so the total mass stays the same.

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5. Inside **an open beaker (no lid)**, hydrochloric acid was added to sodium bicarbonate, and carbon dioxide was produced. What happens to the mass?

Answer **B**

A：Increases　**B** **Decreases**　**C：Stays the same**

> 💡 It gets lighter by exactly the amount of carbon dioxide that **escapes into the air**. The only difference from question 4 is "is there a lid?"

Part 2 **Reasons & mechanism** Think in terms of atoms!

6. Which is the most correct reason why the **Law of Conservation of Mass** holds? **Answer** **A**

A **The combination of atoms changes, but the kinds and numbers of atoms do not change**

B： New atoms are created and old atoms disappear

C： The atoms gradually get smaller during the chemical change

D： No gas is produced in a chemical change

> 💡 A chemical change is a **rearrangement** of atoms. Atoms are **never created or destroyed**, so the kinds and numbers — and therefore the mass — stay the same.

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7. In question 1, what is the **reason the mass increased** when copper was heated in open air? **Answer** **C**

A： The number of copper atoms increased B： It got heavier by the amount of heat that was added

C **Oxygen from the air bonded with the copper**

D： The copper atoms themselves turned into heavier atoms

> 💡 The extra mass is really the **mass of the oxygen that bonded**. Oxygen is invisible, but it definitely has mass.

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8. In question 5, what is the **reason the mass decreased** in the open beaker (no lid)? **Answer** **B**

A： The carbon dioxide disappeared into nothing

B **The carbon dioxide that was produced escaped into the air**

C： All of the hydrochloric acid evaporated D： The number of atoms decreased in the reaction

> 💡 The CO_2 **did not disappear — it just escaped into the air**. If you include the gas that escaped, the mass is conserved.

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9. After the experiment in question 4 (the sealed jar), if you **open the lid** and leave it for a while, what happens to the total mass? **Answer** **B**

A： Increases **B** **Decreases** C： Stays the same

D： Increases at first, then returns to the original

> 💡 When you open the lid, the **CO_2 that was trapped escapes outside**, so the mass decreases by that amount.

Part 3 **Try the calculations** The total is the same before and after!

10. When **10 g** of dilute sulfuric acid was mixed with **17 g** of barium hydroxide solution, **23 g** of barium sulfate and some water formed. How many grams of water formed? **Answer** **B**

A： 2 g **B** **4 g** C： 6 g D： 13 g

> 💡 The total before the reaction is 10 + 17 = **27 g**. The total after must also be 27 g, so the water is 27 − 23 = **4 g**.

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11. Sodium bicarbonate and hydrochloric acid were reacted inside a sealed container. The total before the reaction was **58.0 g**. After the reaction, the lid was opened and after a while it became **57.4 g**. How many grams of carbon dioxide escaped?

Answer **B**

A： 0.4 g　　**B** 0.6 g　　C： 1.6 g　　D： 57.4 g

> 💡 The **mass lost = the mass of the gas that escaped**. 58.0 − 57.4 = **0.6 g**.

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12. **2.0 g** of limestone was put into **10.0 g** of dilute hydrochloric acid (no lid). Gas was produced, and the total after the reaction was **11.2 g**. How many grams of gas were produced?

Answer **A**

A 0.8 g　　B： 1.2 g　　C： 2.0 g　　D： 8.0 g

> 💡 The total before the reaction is 2.0 + 10.0 = **12.0 g**. 12.0 − 11.2 = **0.8 g** is the gas that escaped into the air.

Part 4 True or False

Watch for tricks!

13. "If you dissolve 60 g of sugar in 200 g of water, the total mass will be 260 g even after the sugar dissolves and can no longer be seen."

Answer ○

○ ○ True　　×： × False

> 💡 Even though the sugar **can no longer be seen once it dissolves, its particles have not disappeared**, so the total is exactly 60 + 200 = 260 g. The idea of conservation of mass **also holds for physical changes** such as dissolving or changes of state.

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14. "In a reaction where gas escapes, the Law of Conservation of Mass does not hold."

Answer ×

○： ○ True　　× × False

> 💡 The reading on the scale goes down, but **if you include the gas that escaped, the mass is conserved**. The Law of Conservation of Mass holds for **every chemical change**.

Part 5 **Explain it** ✍️ Write your idea in your own words

15. Even though it is the same reaction (sodium bicarbonate + hydrochloric acid), the mass stayed the same in **the sealed jar** (question 4) but decreased in **the open beaker with no lid** (question 5). Explain why this "difference" occurs.

💬 **Words to use:** **carbon dioxide** | **the air** | **leaves** | **trap**

> 💡 Model answer: In both experiments, carbon dioxide is produced. In **the sealed jar**, the jar **traps** the carbon dioxide inside, so the total mass does not change. In **the open beaker with no lid**, the carbon dioxide **leaves** into **the air**, so the mass decreases by that amount. (The atoms themselves do not disappear, so if you include the gas that escaped, the mass is conserved.)
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> **Marking points**
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> - Mentions that carbon dioxide (gas) is produced in both cases
> - States that sealed = the gas is trapped inside → stays the same
> - States that no lid = the gas leaves into the air → decreases